

Status	Finished
Started	Thursday, 6 November 2025, 1:31 PM
Completed	Thursday, 6 November 2025, 1:59 PM
Duration	28 mins 50 secs
Marks	3.00/3.00
Grade	3.00 out of 3.00 (99.97%)

Information

Information

This page contains all the problems for this test. The very last problem asks you to contact the person in charge of the exam and tell him or her the 4-digit key given in the problem text. In return you will be given a 5-digit signing code which you must give as the answer to the problem.

This problem does not count towards the final score, but **tests missing this code will not count towards the final grade.**

The following rules apply:

- Total time allowed: 30 minutes. The test will automatically close if time runs out.
- UiA's usual rules in regards to cheating on exams apply.

Question 1

Correct

Mark 1.00 out of 1.00

Let A and X be the sets $A = \{\text{a}, \text{b}, \text{c}\}$ and $X = \{!, \%, \&, *, +, -\}$. In this problem we consider words written in the alphabet which is the union of the two sets: $A \cup X$. This means that a word in this alphabet can contain letters from both A and X .

However, not all words are valid. The following rules apply:

1. A word must start with a letter from the set A .
2. A word must end with a letter from the set A .
3. A word can not contain two adjacent letters from the set X .

For example the word **a+baa!c** is valid, but the words **a+baa!c**, **laaa+c** and **acca!** are not.

The length of a word is the number of letters it consists of. For example, the length of **a+baa!c** is 7.

Let a_n equal the number of valid words of length n .

a) Find a recurrence relation satisfied by a_n :

$$a_n - 3a_{n-1} - 18a_{n-2} = 0$$

Your last answer was interpreted as follows: $a_n - 3a_{n-1} - 18a_{n-2} = 0$

The variables found in your answer were: a_{n-2}, a_{n-1}, a_n

Syntax help: To express the symbol " a_n " you should enter **a[n]**. For example, if you think the correct answer is $a_{n+1} - 2a_{n-3} = 1$, you should answer **a[n+1] - 2*a[n-3] = 1**.

b) Compute the value

$$a_1 = 3$$

Your last answer was interpreted as follows: 3

c) Compute the value

$$a_3 = 81$$

Your last answer was interpreted as follows: 81

d) Find the general solution to the recurrence relation satisfied by a_n .

$$a_n = A \cdot 6^n + B \cdot (-3)^n$$

Your last answer was interpreted as follows: $A \cdot 6^n + B \cdot (-3)^n$

The variables found in your answer were: A, B, n

Syntax help: You are free to use constants in your answer. For example, if you think the correct answer is $2n - A \cdot 3^n + B$, you should answer **2*n - A*3^n + B**.

Question 2

Correct

Mark 1.00 out of 1.00

In this problem you are tasked with computing with elements in the symmetric group of permutations, $((S_{\{5\}}, \cdot))$.

Syntax guide: The answer to each task is an element of the group $(S_{\{5\}})$ and should be given as a (2×5) matrix. E.g. if you think the answer to a question is the permutation denoted by $\left[\begin{matrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 3 & 1 & 5 & 4 \end{matrix} \right]$ then you should answer

1	2	3	4	5
2	3	1	5	4

a)

$$\left(\left[\begin{array}{ccccc} 1 & 2 & 3 & 4 & 5 \\ 4 & 2 & 1 & 3 & 5 \end{array} \right] \cdot \left[\begin{array}{ccccc} 1 & 2 & 3 & 4 & 5 \\ 5 & 4 & 3 & 1 & 2 \end{array} \right] \right) = \left[\begin{array}{ccccc} 1 & 2 & 3 & 4 & 5 \\ 1 & 4 & 5 & 3 & 2 \end{array} \right]$$

Your last answer was interpreted as follows: $\left[\begin{array}{ccccc} 1 & 2 & 3 & 4 & 5 \\ 1 & 4 & 5 & 3 & 2 \end{array} \right]$

b)

$$\left(\left[\begin{array}{ccccc} 1 & 2 & 3 & 4 & 5 \\ 2 & 4 & 3 & 1 & 5 \end{array} \right] \cdot \left[\begin{array}{ccccc} 1 & 2 & 3 & 4 & 5 \\ 2 & 4 & 1 & 5 & 3 \end{array} \right] \right) = \left[\begin{array}{ccccc} 1 & 2 & 3 & 4 & 5 \\ 4 & 5 & 1 & 2 & 3 \end{array} \right]$$

Your last answer was interpreted as follows: $\left[\begin{array}{ccccc} 1 & 2 & 3 & 4 & 5 \\ 4 & 5 & 1 & 2 & 3 \end{array} \right]$

c)

$$\left(\left[\begin{array}{ccccc} 1 & 2 & 3 & 4 & 5 \\ 4 & 2 & 1 & 3 & 5 \end{array} \right] \right)^{-1} = \left[\begin{array}{ccccc} 1 & 2 & 3 & 4 & 5 \\ 3 & 2 & 4 & 1 & 5 \end{array} \right]$$

Your last answer was interpreted as follows: $\left[\begin{array}{ccccc} 1 & 2 & 3 & 4 & 5 \\ 3 & 2 & 4 & 1 & 5 \end{array} \right]$

d)

$$\left(\left[\begin{array}{ccccc} 1 & 2 & 3 & 4 & 5 \\ 5 & 4 & 3 & 1 & 2 \end{array} \right] \right)^{-1} = \left[\begin{array}{ccccc} 1 & 2 & 3 & 4 & 5 \\ 4 & 5 & 3 & 2 & 1 \end{array} \right]$$

Your last answer was interpreted as follows: $\left[\begin{array}{ccccc} 1 & 2 & 3 & 4 & 5 \\ 4 & 5 & 3 & 2 & 1 \end{array} \right]$

Question 3

Correct

Mark 1.00 out of 1.00

a) Let $G = (\mathbb{Z}/2\mathbb{Z})^{\times 4}$, and let $H \subset G$ be the subgroup $H = \{ \left(\begin{smallmatrix} 0 \\ 0 \\ 0 \\ 0 \end{smallmatrix} \right), \left(\begin{smallmatrix} 0 \\ 0 \\ 0 \\ 1 \end{smallmatrix} \right), \left(\begin{smallmatrix} 0 \\ 0 \\ 1 \\ 0 \end{smallmatrix} \right), \left(\begin{smallmatrix} 0 \\ 0 \\ 1 \\ 1 \end{smallmatrix} \right), \left(\begin{smallmatrix} 0 \\ 1 \\ 0 \\ 0 \end{smallmatrix} \right), \left(\begin{smallmatrix} 0 \\ 1 \\ 0 \\ 1 \end{smallmatrix} \right), \left(\begin{smallmatrix} 0 \\ 1 \\ 1 \\ 0 \end{smallmatrix} \right), \left(\begin{smallmatrix} 0 \\ 1 \\ 1 \\ 1 \end{smallmatrix} \right) \}$.

What is the index of $H \subset G$?

$[G:H] =$

Your last answer was interpreted as follows: $[2]$

b) Let D_4 be the dihedral group of order 8 of symmetries of the unit square centered at the origin of $\mathbb{R}^2$. Let $\sigma \in D_4$ be the symmetry that rotates -180 degrees around the origin.

Compute the index of the subgroup generated by σ :

$[D_4 : \langle \sigma \rangle] =$

Your last answer was interpreted as follows: $[4]$

Question 4

Incorrect

Mark 0.00 out of 0.00

Signing code

Before closing the test you must answer this problem with a signing code given to you by the person in charge of the test.

Tests missing this signing code will be ignored and will not count towards the final score.

Key: $\{220\}$

Signing code:

Your last answer was interpreted as follows: $[53689]$

◀ Test 2 (topics 4-6: Languages, Machines, Regular expressions, Correct programs, Relations)

Jump to...

